

UNIVERSITY OF ESSEX

Postgraduate Examinations 2019

MACHINE LEARNING AND DATA MINING

Time allowed: **TWO** hours

Candidates are permitted to bring into the examination room:

Calculator – Casio FX-83GT PLUS or Casio FX-85GT PLUS only

The paper consists of **THREE** questions.

Candidates must answer **ALL** questions.

The questions are **not** of equal weight.

Questions 1 and 2 are worth 35% of the marks for the paper each.

Question 3 is worth 30% of the marks for the paper.

The percentages shown in brackets provide an indication of the proportion of the total marks for the **PAPER** that will be allocated for that part of the question.

**Please do not leave your seat unless you are given permission by an invigilator.
Do not communicate in any way with any other candidate in the examination room.**

Do not open the question paper until told to do so.

All answers must be written in the answer book(s) provided.

All rough work must be written in the answer book(s) provided. A line should be drawn through any rough work to indicate to the examiner that it is not part of the work to be marked.

At the end of the examination, remain seated until your answer book(s) have been collected and you have been told you may leave.

Question 1

- (a) Support vector classifiers can be thought of as instance-based learners. Explain why, indicate what similarity measures they use, explain what instances they retain and how these contribute to the class prediction. [8%]
- (b) Figure 1.1 (below) shows a set of training examples and the decision function of a non-linear support vector classifier that was trained on these data. [9%]
- Identify which of the training points are support vectors;
 - Describe qualitatively what happens to the decision boundary and to the margin if a new training example of class “+1” and features $[1.0, 1.0]$ is added to the training set.

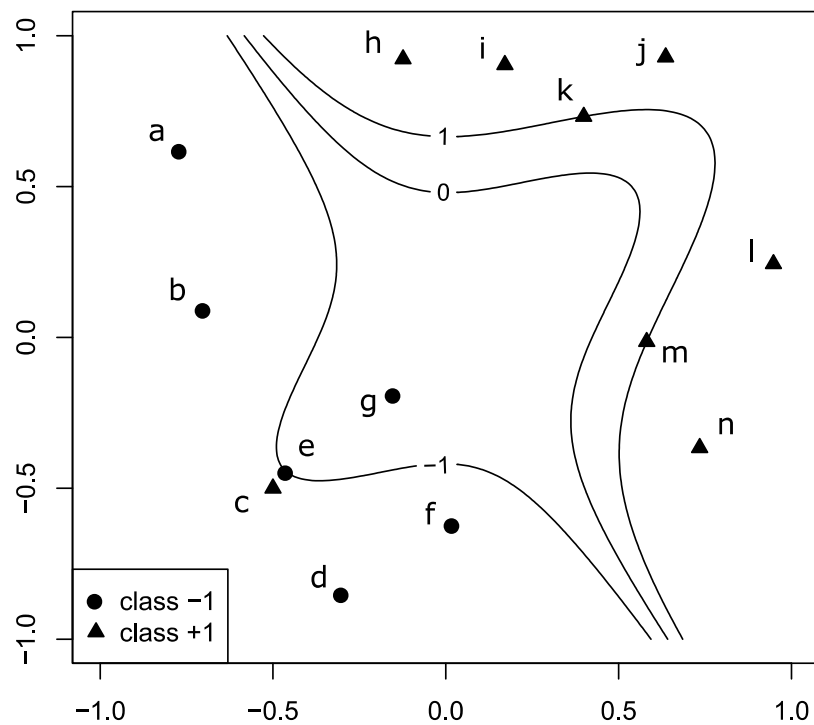


Figure 1.1

Question 1 continues...

Question 1 (continued)

- (c) Consider the radial basis function kernel $K_{\text{RBF}}(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \cdot \|\mathbf{x} - \mathbf{z}\|^2)$. Sketch the graph of the value of the kernel as a function of $d = \|\mathbf{x} - \mathbf{z}\|^2$ (i.e. the squared distance between the two points $\mathbf{x}$ and $\mathbf{z}$) for a small and a large value of the parameter γ . Then draw a parallel between γ and the parameter k in k-NN classifiers and explain the effect they have on the behaviour of the induced classifiers. In particular, describe how large or small values of γ and k may lead to underfitting or overfitting in SVMs and k-NNs, respectively. [8%]
- (d) Draw a sketch of the *0-1 loss* and of the *hinge loss* functions. Briefly explain three disadvantages of the *0-1 loss* compared to the *hinge loss* when building linear classifiers. [10%]

Question 2

- (a) Briefly explain the difference between supervised and unsupervised learning. Then list two examples of supervised learning algorithms and two examples of unsupervised learning algorithms. [10%]
- (b) Use the k-means algorithm with 2-norm (Euclidean) distance to cluster the following 6 examples into 2 clusters: $p=(-3,2)$, $q=(1,2)$, $r=(11,8)$, $s=(7,8)$, $t=(-1,11)$, $u=(9,-1)$. Suppose that the initial seeds (centres of each cluster) are the instances t and u . Run the k-means algorithm for three iterations or until convergence, whichever comes first. For each iteration show the cluster assignments (which instances belong to each cluster) and the new centres of the clusters. [12%]
- (c) Describe two limitations of the k-means clustering algorithm. [5%]
- (d) An outlier is an instance that is very far from all other points. Outliers can negatively affect the result of a machine learning algorithm. A possible solution is to try to identify and remove outliers before applying the algorithm while another possibility is to use an algorithm which is more robust to outliers. Using your intuition and accumulated knowledge of machine learning, devise and describe a possible modification to the k-means clustering algorithm to make it more robust to outliers. [8%]

Question 3

- (a) You are told that a given system achieves 97% accuracy on a binary (2 classes) classification task. What additional pieces of information do you need to know before you can decide whether this is a good result? List and discuss at least two. [8%]
- (b) Testing a classifier's performance using a 10-fold cross-validation scheme produced the following confusion matrix (Table 3.1): [12%]

		Predicted	
		Yes	No
Actual	Yes	72	18
	No	8	32

Table 3.1

Indicate the values of TP, FP, TN, FN from the matrix, then compute the following measures:

- (i) Estimated prior probability of class Yes
 - (ii) Recall (also known as true positive rate or sensitivity)
 - (iii) Precision (also called positive predictive value)
 - (iv) Classification accuracy
- (c) Suppose that several different learning procedures have been applied to a classification task and that each has achieved partial success. Describe, in detail, a learning method that could be used to enable all the learning procedures to co-operate and make more accurate predictions. [10%]

END OF PAPER CE802-7-AU